Integrating Local–Global Structural Interaction Using Siamese Graph Neural Network for Urban Land Use Change Detection From VHR Satellite Images

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***Abstract*— Detecting land use changes in urban areas from very-high-resolution (VHR) satellite images presents two primary challenges: 1) traditional methods focus mainly on comparing changes in land cover-related features, which are insufficient for detecting changes in land use and are prone to pseudo-changes caused by illumination differences, seasonal variations, and subtle structural changes and 2) spatial structural information, which is characterized by topological relationships among land cover objects, is crucial for urban land use classification but remains underexplored in change detection. To address these challenges, this study developed a local–global structural interaction network (LGSI-Net) based on a Siamese graph neural network (SGNN) that integrates high-level structural and semantic information to detect urban land use changes from bitemporal VHR images. We developed both local structural feature interaction module (LSIM) and global structural feature interaction module (GSIM) to enhance the representation of bitemporal structural features at the global scene graph and local object node levels. Experiments on the publicly available MtS-WH dataset and two generated datasets, LUCD-FZ and LUCD-HF, show that the proposed method outperforms the existing bag of visual word (BoVW)-based method and CorrFusionNet. Furthermore, we evaluated the detection performance for different semantic feature extraction strategies and structural feature extraction backbones. The results demonstrate that the proposed method, which integrates high-level semantic and graph isomorphism network (GIN)-derived structural features achieves the best performance. The method trained on the LUCD-FZ dataset was successfully transferred to the LUCD-HF dataset with different urban landscapes, indicating its effectiveness in detecting land use changes from VHR satellite images, even in areas with relatively large imbalances between changed and unchanged samples.**

***Index Terms*— Local–global structural interaction, Siamese graph neural networks (SGNNs), urban land use change detection, very-high-resolution (VHR) satellite images.**

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I. Introduction

LAND use information reveals the use of Earth's surface and provides valuable insights into human–environment interactions [1]. With rapid urbanization, detailed information on changes in urban land use is essential for developing appropriate strategies, promoting rational infrastructure planning, and ensuring efficient resource utilization [2], [3], [4], [5]. Previously, data on urban land use changes were mainly acquired through visual interpretation and field surveys because of limited data availability and insufficient spatial resolution. However, advancements in remote sensing technology have made remote sensing imaging a key technique for obtaining data on urban land use changes. Among them, very-high-resolution (VHR) satellite images are effective for identifying detailed land use types (e.g., residential, commercial, and industrial land) and detecting changes in land use [6], [7], [8], [9], [10].

Many methods, including pixel-, object-, and scene-based methods, according to the granularity of the basic spatial analysis units, have been employed to detect land changes from VHR satellite images [11], [12], [13]. Pixel-based change detection methods adopt individual pixels as the basic units and compare pixels across identical locations over time based on predefined thresholds [14], [15], [16]. These methods are effective in extracting spectral and textural features and are commonly used for moderate- or low-resolution images. Object-based methods group similar and adjacent pixels into image objects and compare bitemporal feature differences at the object level [17], [18], [19], [20]. These methods can capture rich geometric features and are more suitable for analyzing VHR images in urban settings. Scene-based methods, which have recently attracted attention, use image scenes as the basic analysis unit [21], [22]. Image scenes encompass a large spatial neighborhood allowing for the extraction of high-level semantic features necessary for detecting functional land use changes [23], [24]. For instance, as illustrated in Fig. 1, while pixel- and object-based methods can identify physical changes such as buildings or roads in industrial areas, these might not reflect functional changes. Scene-based methods, however, can detect such functional changes by considering the broader spatial context of land use.

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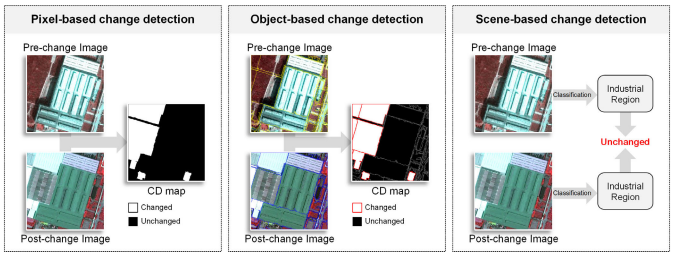


Fig. 1. Demonstration of the differences between pixel-, object-, and scene-based change detection.

Therefore, in this study, we employed image scenes as the basic units for detecting changes in urban land use.

Scene-based image analysis, which involves scene classification and change detection, has been widely used in VHR image processing [25], [26], [27], [28]. Traditional scene-based change detection involves extracting various image features based on feature engineering or computer vision feature descriptors and then applying a change detector to identify changed areas based on the differences in bitemporal features. Wu et al. [26] employed the bag of visual word (BoVW) model and kernel slow feature analysis to improve scene change detection, and Du et al. [29] developed an automatic framework that combines latent Dirichlet assignment and multivariate alteration detection for identifying scene changes. However, these traditional methods face significant challenges in urban areas due to the complex composition and structure of land cover objects within scenes. This complexity complicates the extraction of effective features for characterizing urban land use, thereby increasing the difficulty of detecting changes in urban land use from bitemporal images.

Recent studies have shown that deep learning methods, which can learn hierarchical representations, outperform traditional computer vision techniques, such as BoVW, in remote sensing image analysis, such as change detection [30], [31], [32], [33], image super-resolution mapping [34], [35], [36], and semantic segmentation [37]. For scene-based change detection, Ru et al. [38] employed a convolutional neural network (CNN) with deep canonical correlation analysis to extract semantic features and model temporal correlations from image pairs. Fang et al. [39] developed a framework that combines a differential aggregation network and a class probability-based fusion strategy to enhance feature representation and improve accuracy. Li and Stein [40] reported that analyzing the spatial composition and structure of land cover elements can aid in the inference of urban land use types. However, the effectiveness of spatial structure information of land cover elements in urban land use change detection has not been sufficiently explored, which motivates this study.

The spatial structure of land cover objects can be quantified by modeling the topological relationships between neighboring land cover objects within an image scene. Previous studies have shown that analyzing such structural relationships enhances change detection accuracy [40]. Based on graph theory, land cover objects and their interactions can be represented as nodes and edges in a graph [41], [42], [43], [44], [45], [46]. Graph neural networks (GNNs) are suitable for automatically deriving high-level structural properties from graph-structured data owing to their exceptional ability to capture complex structural features. In addition, GNNs can integrate attribute information associated with the nodes and edges of a graph, making them suitable for graph learning tasks. Chen et al. [47] employed a graph convolutional network (GCN) to learn representative structural features and combined it with postprocessing operations to detect object-level changes. Wu et al. [48] developed a novel multiscale GCN-based method that captures structural features at various scales, aggregating them with a fusion module to enhance change detection. These studies successfully extracted high-level structural features from image-derived graph data using GNNs. However, they did not account for the temporal correlation between bitemporal image pairs, resulting in the detection of pseudo-changes caused by external factors.

In this study, we employed bitemporal VHR satellite images to detect changes in urban land use at the scene level, exploiting the spatial structures of land cover objects using graph-based methods. We developed a Siamese GNN (SGNN) equipped with an attention-based global structural feature enhancement module (GSEM) that dynamically adjusts attention scores for each land cover object as the structural feature extractor to derive high-level structural features from bitemporal image scene graphs. Previous studies have shown the importance of feature interactions in capturing temporal differences between bitemporal image pairs, providing discriminative information for change detection [49], [50], [51]. Based on this, we designed two bitemporal structural feature interaction modules that operate at both the global (scene graph) and local (object node) levels. The global graph-level interaction module captures the overall relationships, and the local node-level interaction module provides fine-grained details. The key contributions of this study are as follows.

1) A novel framework, the local–global structural interaction network (LGSI-Net), is proposed for urban land use change detection from bitemporal VHR images by modeling the spatial structure information of land cover objects. Based upon a Siamese architecture, it employs a GNN extractor to derive high-level structural features of adjacent land cover objects and identifies land use changes by analyzing structural differences.
2) A global structural feature interaction module (GSIM) based on neural tensor networks (NTNs) is developed to capture the temporal correlation of bitemporal features at the global graph level, thereby reducing false detection caused by changes in external factors. Moreover, a local structural feature interaction module (LSIM) is proposed to derive fine-grained correlation representations of bitemporal node-level structural features. It minimizes false positives and missed detections caused by local structural variations.
3) Two new land use change detection datasets, named LUCD-FZ and LUCD-HF, are generated to evaluate the proposed method and to facilitate the development of more advanced techniques in land use change detection (available at: https://github.com/Imp83121/LUCD_Datasets).

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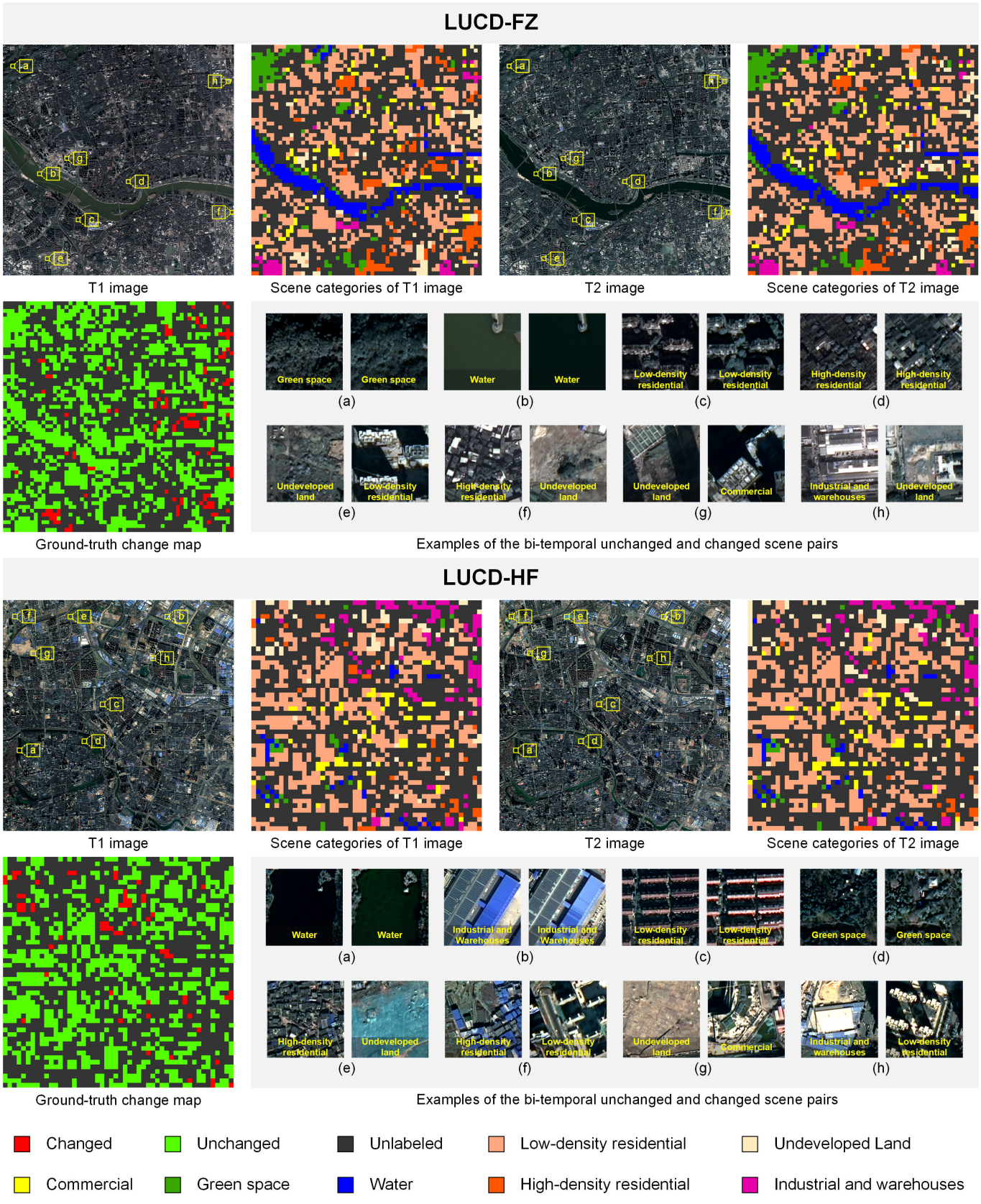


Fig. 2. Overview of the created LUCD-FZ and LUCD-HF datasets.

The remainder of this article is organized as follows. Section II describes the study areas and datasets. Section III illustrates the proposed method for urban scene-based land use change detection. The experimental results and related analysis are presented in Section IV, followed by a discussion in Section V and conclusions of this article in Section VI.

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TABLE I
DETAILED INFORMATION OF THE MTS-WH, LUCD-FZ, AND LUCD-HF DATASETS

Datasets	Phases	Sensors	Resolution (m)	Dates	Sizes (pixels)	Areas (km^2)
MtS-WH	T1	IKONOS	1.0	2002	7200×	43.20
	T2			2009	6000	
LUCD-FZ	T1	GaoFen-2	0.8	2016.12.07	12000×	92.16
	T2			2020.02.18	12000	
LUCD-HF	T1	GaoFen-2	0.8	2017.01.26	10000×	64.00
	T2			2021.12.20	10000	

II. STUDY AREAS AND DATASETS

We tested the proposed method using three VHR remote sensing datasets: the publicly available MtS-WH dataset in Wuhan, China [26], [28], and two newly created datasets in Fuzhou, China, and Hefei, China, namely, LUCD-FZ and LUCD-HF (see Fig. 2).

Table I lists detailed information of the three datasets used in this study. The MtS-WH dataset [26], [28], an early public dataset for scene-based change detection, consists of two optical images acquired from the IKONOS satellite, covering an area of the Hanyang district in Wuhan. The images were divided into nonoverlapping scenes of 150 × 150 pixels and annotated with eight classes: parking, water, sparse houses, dense houses, residential region, idle region, vegetation region, industrial region, and unlabeled scenes. The unlabeled category refers to the scene images that cannot be accurately classified by visual interpretation [26], [39]. The covered area of this dataset is characterized by various land use types and has experienced rapid development, resulting in evident land use changes. Both the LUCD-FZ and LUCD-HF datasets consist of two optical images acquired from the Chinese Gaofen-2 (GF2) satellite. The GF2 images were captured by a PMS1 sensor having four multispectral bands with a spatial resolution of 2.5 m and one panchromatic band of 0.78 m. We performed pansharpening operations to fuse multispectral and panchromatic bands for each image and resampled the pansharpened images to 0.8 m for experimental convenience. The preprocessed images were then divided into nonoverlapping scene images with 200 × 200 pixels and annotated with seven classes: high-density residential, low-density residential, commercial, industrial, and warehouses, green space, water, and undeveloped land. The ground-truth scene labels were determined via visual interpretation and points of interest from Baidu Maps. The classification system was adapted from the Chinese National Land Use Classification Standard (GB/T 21010-2017) and tailored for analyzing land use in urban areas [1], [8], [40], [52]. Here, we aggregated natural land use classes, i.e., forest, orchard, and grass into a single category, i.e., green space. Compared to the MtS-WH dataset, the LUCD-FZ and LUCD-HF datasets exhibit a stronger imbalance between changed and unchanged samples. Such imbalance also reflects the complexity of current urban dynamics, making it suitable for evaluating the effectiveness of our method for land use change detection in urban areas under realistic conditions.

III. METHODOLOGY

Fig. 3 shows the workflow of the proposed detection of urban land use changes using LGSI-Net. We modeled the topological relationships between neighboring land cover objects initially derived by superpixel segmentation on stacked bitemporal images as graph-structured data. Then, we employed a pretrained semantic segmentation model to extract high-level semantic features as node attributes. Next, we constructed a weight-sharing SGNN complemented by an attention-based GSEM as the structural feature extractor, enabling the extraction of high-level structural features while seamlessly integrating the semantic information of graph nodes. Furthermore, we incorporated the GSIM module, which operates at the graph level to reduce false detections caused by changes in external factors, and the LSIM module to derive fine-grained node-level correlations, minimizing false positives and missed detections due to local structural variations. Finally, we used the enhanced bitemporal structural features to detect changes in land use.

A. Characterizing the Topological Relationships Between Land Cover Objects Into Graph-Structured Data

Given a pair of scene images acquired at times T_1 and T_2, we denote the T_1 and T_2 images $X_1 \in \mathbb{R}^{H\times W\times C}$ and $X_2 \in \mathbb{R}^{H\times W\times C}$, respectively, where H, W, and C are the height, width, and number of channels of the images, respectively. The pixels of the images are denoted as $x_1(h, w, c)$ and $x_2(h, w, c)$, respectively.

1) Generation of Graph-Structured Data: As shown in Fig. 3(a), the generation of graph-structured data begin with superpixel segmentation of images X_1 and X_2 to generate basic image objects for analysis. Following the report by Chen et al. [47], we adopted a co-segmentation strategy, which stacks the two images and used the fractal net evolution approach (FNEA) [53] for segmentation. Let $\Omega = \{\Omega_1, \ldots, \Omega_{N_{\text{lc}}}\}$ be the segmentation result, where N_{lc} is the number of image objects. The ith object of X_1 and X_2 is expressed as $O_i^{X_1} = \{X_1(h, w, :) \mid h, w \in \Omega_i\}$ and $O_i^{X_2} = \{X_2(h, w, :) \mid h, w \in \Omega_i\}$, respectively. Based on the generated image objects, we used the graph theory to represent the topological relationships between neighboring image objects. Let $G_1 = (V_1, E_1, A_1)$ represent the graph of prechange image X_1 consisting of a set of nodes $V_1 = \{v_1, \ldots, v_n\}$ and a set of edges $E_1 = \{e_1, \ldots, e_m\}$, where n and m are the number of nodes and edges, respectively. A node v_i refers to the image object Ω_i and an edge e_i refers to the spatial adjacency relationship between two objects. The adjacency matrix A measures the relationship between image objects as follows:

$$A_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $A_{ij} = 1$ indicates that the two nodes v_i and v_j are connected; otherwise, $A_{ij} = 0$. This representation encodes the topological structure of the graph G_1, providing a comprehensive overview of how image objects are spatially related to each other, enabling the inference of potential pattern within

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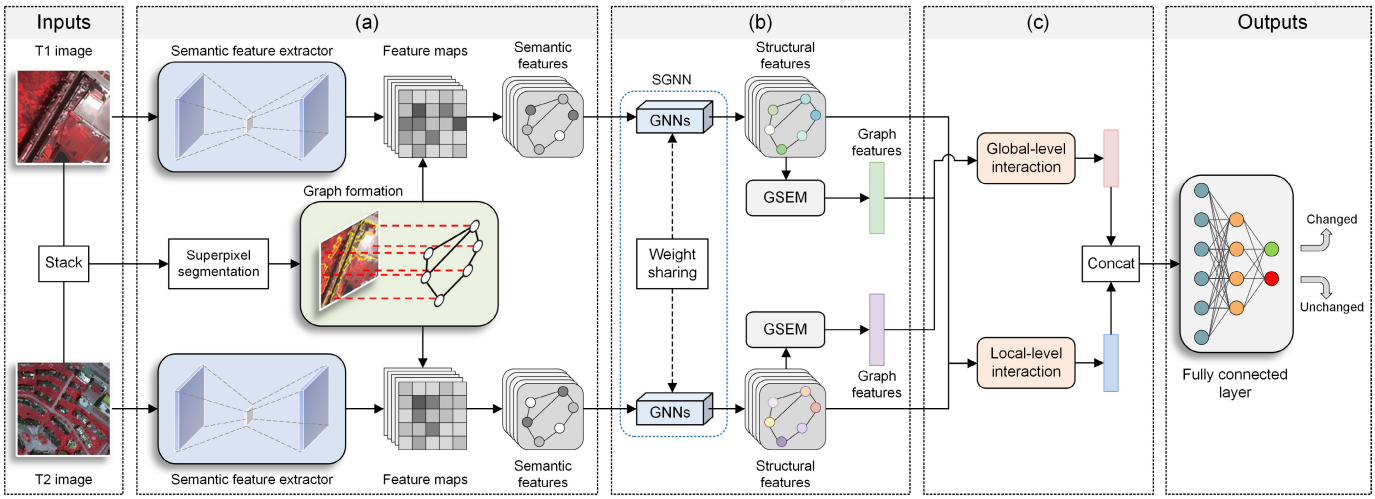


Fig. 3. Workflow of the proposed urban land use change detection. It consists of the (a) construction of graph-structured data from bitemporal VHR scene images, including the extraction of node features using a semantic feature extractor, (b) extraction of high-level structural feature using weight-sharing SGNN, and (c) interaction of local–global bitemporal structural features.

the image scene. Similarly, we constructed the graph G_2 for the postchange image X_2. Besides, features F_{X_1} and F_{X_2} can be, respectively, assigned to each node of G_1 and G_2.

2) Extraction of High-Level Node Semantic Features: To describe node attributes, we extracted high-level semantic features that characterize land cover classes. Thus, the segmented image objects are tagged with land cover information, referred to as land cover objects. Furthermore, we distinguished seven urban land cover classes: trees, grass, shadow, building, water, bare land, and others. We used the recently developed CLCFormer [37], a semantic segmentation network that integrates CNN and Transformer architectures, to extract land cover-related semantic features. CLCFormer is highly effective in land cover classification from remote sensing images. Unlike conventional feature engineering methods, this method exhibits automatic learning capabilities and enhances contextual awareness. The main parameters of the CLCFormer were configured as follows: the batch size was set to 8, the number of epochs was set to 25, the learning rate was set to 0.0001, and Adamw was selected as the optimizer. The training dataset was collected by digitizing the VHR images. For image X_1, we extracted 32 feature maps denoted as $M_{X_1} \in \mathbb{R}^{H\times W\times C_M}$, $(C_M = 1, \ldots, 32)$ through the pretrained CLCFormer feature extractor. To initially characterize node features F_{X_1} for a prechange graph G_1, we combined high-level semantic information about the feature maps M_{X_1} and spatial information about the segmentation result Ω by aggregating pixel-level features within each image object. Similarly, we obtained the node features F_{X_2} for postchange graph G_2. High-level structural features were extracted from bitemporal graph-structured data and were used to detect changes in land use.

B. Structural Feature Extraction Based on GNNs

Previous studies have shown the significance of the spatial arrangement and structure of land cover objects in distinguishing different land use types [40]. Herein, scenes are treated as basic land use units because they represent larger spatial scales and often encompass more complex semantic information. Solely relying on changes in land cover object attributes is insufficient for the accurate detection of changes in land use. Thus, we employed GNNs to extract high-level structural features from scene graphs, thereby enhancing the interpretation of land use change information. However, conventional single-branch GNNs are limited to modeling the temporal correlations of land cover objects between bitemporal graphs. Therefore, we constructed an SGNN that extracts bitemporal structural features and captures their structural difference from the graph data (i.e., G_1 and G_2) derived from bitemporal VHR images. Fig. 4 shows the architecture of the proposed SGNN. It comprises a Siamese architecture with two subnetwork GNNs connected through weight sharing. This enhances the representation of spatial relationships and structural correlations between bitemporal features and improves the model's generalizability. We selected the graph isomorphism network (GIN) [54] as the subnetwork because of its remarkable performance in various graph-related tasks [55], [56]. Its robustness and superior ability to capture complex structural patterns and relationships in graph-structured data make it suitable for detecting changes. Besides, GIN exhibits efficient implementation and scalability, making it computationally feasible for large-scale applications and ensuring practical utility in real-world scenarios.

1) SGNN Architecture: SGNN takes the bitemporal graph structure (i.e., G_1 and G_2) as input and learns the hidden structural information embedded in both graphs through several weight-sharing stacked GNN layers. This process can be expressed as follows:

$$\begin{cases} S_1 = \mathrm{GNN}^{(L)}\left(\mathrm{GNN}^{(L-1)}, \ldots, \mathrm{GNN}^{(1)}(G_1)\right) \\ S_2 = \mathrm{GNN}^{(L)}\left(\mathrm{GNN}^{(L-1)}, \ldots, \mathrm{GNN}^{(1)}(G_2)\right) \end{cases} \tag{2}$$

where L represents the number of GNN layers, and $S_1 \in \mathbb{R}^{N_{\mathrm{lc}}\times C_S}$ and $S_2 \in \mathbb{R}^{N_{\mathrm{lc}}\times C_S}$ are the extracted high-level structural features with C_S channels.

In GNNs, the number of layers significantly affects the expressive capacity of a model. On the one hand, insufficient

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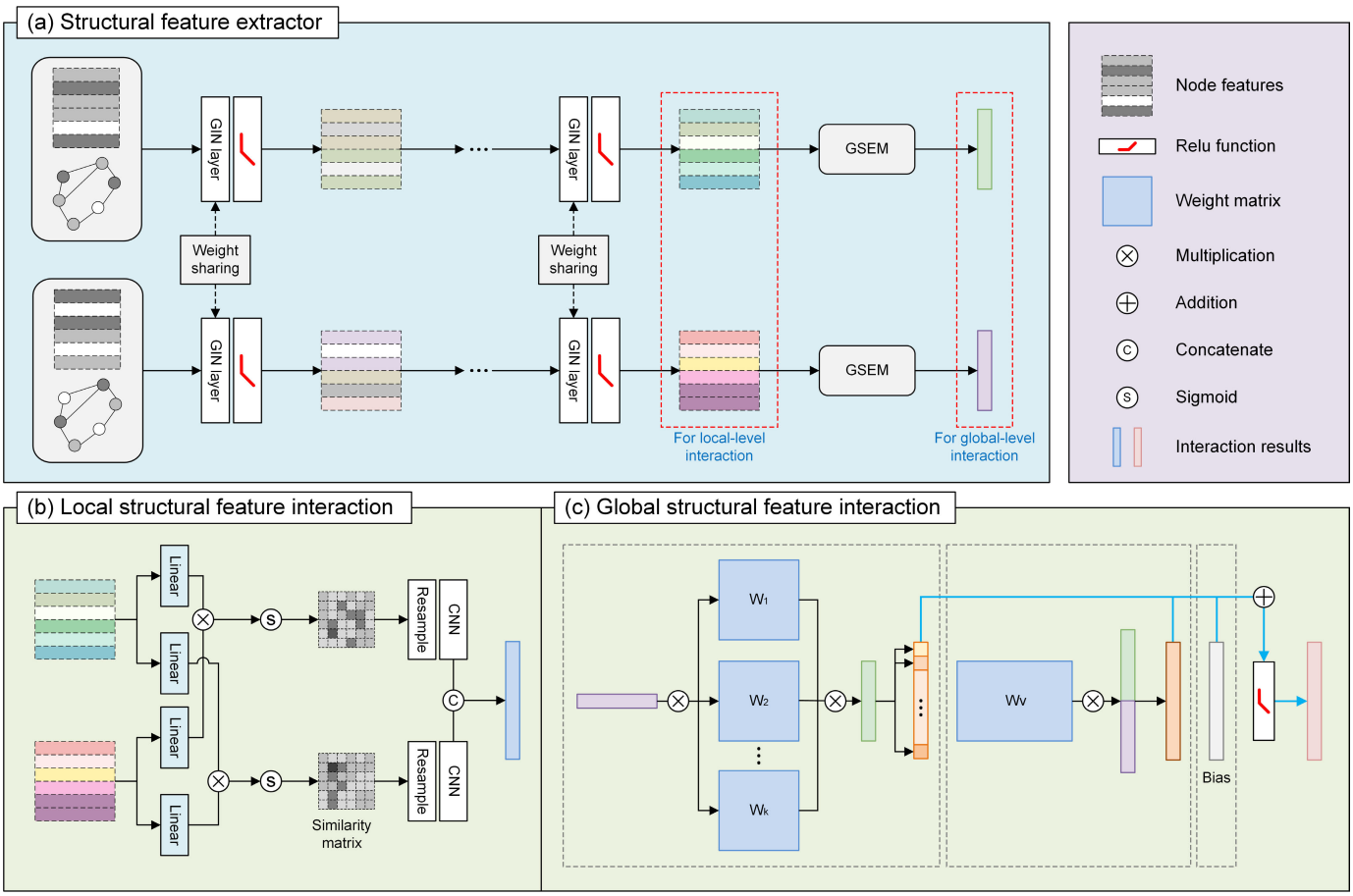


Fig. 4. Details of (a) structural feature extractor using SGNN complemented by GSEM module, (b) LSIM, and (c) GSIM.

layers may constrain the ability of the model to capture complex patterns. Conversely, excess layers can cause problems, such as feature smoothing or overfitting. Therefore, it is important to carefully consider the layer count to balance between expressive power and model performance. In this study, after evaluating GNN architectures with varying layer depths, we found that a three-layer configuration is optimal for extracting structural information from land cover.

2) Graph Isomorphism Network: GNNs exploit graph-structured data to learn the representation vectors h_v for a node or h_g for the entire graph. Herein, we employed the neighborhood aggregation strategy, where the representation of a node is updated iteratively by aggregating information from its neighbors. The kth layer of GNN is expressed as

$$a_v^{(k)} = \text{AGGREGATE}^{(k)}\left(h_u^{(k-1)} : u \in N(v)\right) \quad (3)$$

$$h_v^{(k)} = \text{COMBINE}^{(k)}\left(h_v^{(k-1)}, a_v^{(k)}\right) \quad (4)$$

where $h_v^{(k)}$ is the feature vector of node v at the kth iteration/layer, and $h_v^{(k-1)}$ is the feature vector of node v from the previous $(k-1)$th layer. We set the initial value of $h_v^{(0)}$ as node high-level semantic features and $N(v)$ represents the set of nodes adjacent to v. The selection of $\text{AGGREGATE}^{(k)}(\cdot)$ and $\text{COMBINE}^{(k)}(\cdot)$ plays a crucial role in determining how effectively the network learns and represents the inherent graph structural features. For $\text{AGGREGATE}^{(k)}(\cdot)$, although the original GIN utilized the *add* aggregation function, we used the elementwise *mean* aggregation because of the complexity of semantic information for land cover nodes. This aggregation method effectively captures the comprehensive information while minimizing the impact of noise, thereby enhancing robustness against outliers and ensuring a stable graph representation. For $\text{COMBINE}^{(k)}(\cdot)$, we followed GIN's approach and employed a multilayer perceptron (MLP) to model and update node information. This approach enhances the expressive capabilities of GIN, thereby enabling them to capture and represent complex structural patterns in graph-structured data. The AGGREGATE and COMBINE steps are integrated as follows:

$$h_v^{(k)} = \text{MLP}^{(k)}\left(\frac{\sum_{u\in N(v)} h_u^{(k-1)}}{|N(v)|} + (1+\epsilon^{(k)}) \cdot h_v^{(k-1)}\right) \quad (5)$$

where $\text{MLP}^{(k)}$ represents an MLP specific to the kth layer, $\epsilon^{(k)}$ is a learnable parameter, introducing adaptability to changes in the graph structure, $(1/|N(v)|)\sum_{u\in N(v)} h_u^{(k-1)}$ involves aggregating the feature vectors of neighboring nodes u in the $(k-1)$th layer, and $h_v^{(k)}$ is the high-level structural features extracted by the kth GIN.

3) Global Structural Feature Enhancement via an Attention Mechanism: We derived global graph-level structural features by aggregating local node-level structural features extracted by the SGNN. To enhance the representation of global features, we designed an attention-based GSEM (Fig. 5), which allows

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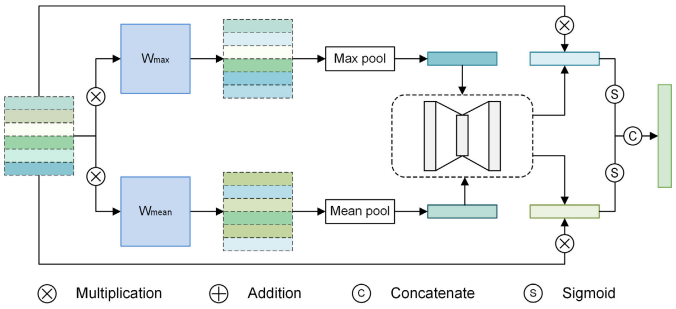


Fig. 5. Illustration of the GSEM module.

the model to obtain comprehensive global structural features guided by the inherent importance of individual land cover nodes.

Given the prechange node-level structural features $S_1 \in \mathbb{R}^{N_{\text{lc}} \times C_S}$, two distinct attention context descriptors, which represent the average and maximum pooled attention contexts, were computed by multiplying two weight matrices to the node feature matrix and then averaging and taking the maximum value of the resulting products. These two contexts provide global structural and feature information about a graph that is adaptive to the given similarity metric, by learning the weight matrix. Each attention context descriptor is further processed through a shared network composed of an MLP layer to obtain the resulting node attention scores. The average and maximum node attention scores are then individually multiplied by the original node features to generate global structural features using maximum and average pooling, corresponding to $S_{1\max}$ and $S_{1\text{avg}}$. By concatenating the results, the final global graph-level feature representation S_{g_1} can derived and the GSEM is calculated as follows:

$$S_{1\max} = \sigma(\text{MLP}(\text{MaxPool}(S_1 W_1)) \cdot S_1) \tag{6}$$

$$S_{1\text{avg}} = \sigma(\text{MLP}(\text{AvgPool}(S_1 W_2)) \cdot S_1) \tag{7}$$

$$S_{g_1} = \text{Concat}\left(S_{1\max}, S_{1\text{avg}}\right) \tag{8}$$

where $W_1 \in \mathbb{R}^{C_S \times C_S}$ and $W_2 \in \mathbb{R}^{C_S \times C_S}$ denote the learnable weight matrices, and σ is the Sigmoid function. The resulting vector S_{g_1} represents the global structural feature of the graph.

C. Local–Global Bitemporal Structural Feature Interaction

To minimize false detection due to changes in illuminance conditions and seasonal variations in the scene-based detection of land use changes, we developed a GSIM using NTNs [57] to capture the temporal correlation of global structural features. Furthermore, to minimize false positives and missed detections of land use changes caused by local structural variations in scenes containing complex internal land cover features, we developed an LSIM to provide detailed change information about the scene images at a land cover object level. This approach captures both coarse- and fine-grained change information, facilitating precise analysis and detection of land use changes.

1) Global Structural Feature Interaction Module: Given the global structural features S_{g_1} and S_{g_2} of two scene graphs representing the scenes before and after a change, their relationship can be modeled using straightforward operations, such as subtraction. However, this approach may lead to insufficient or weak interactions between two vectors due to limitations in the expressive power and parameterization of graph embedding representations [57]. To address this challenge, we employed NTN to model the relationship between bitemporal global structural features while exploring their elementwise dependence. NTN is defined as

$$I_{\text{global}} = f\left(S_{g_1}^{\text{T}} W^{[1:K]} S_{g_2} + V \begin{bmatrix} S_{g_1} \\ S_{g_2} \end{bmatrix} + b\right) \tag{9}$$

where $W^{[1:K]} \in \mathbb{R}^{C_S \times C_S \times K}$ denotes the learnable weight matrices, $[\cdot]$ denotes the concatenation operation, $V \in \mathbb{R}^{K \times 2C_S}$ is a weight tensor, $b \in \mathbb{R}^K$ is a bias vector, and $f(\cdot)$ is an activation function. K is a hyperparameter controlling the number of interaction scores produced by the model for each graph embedding pair. In this study, K is set to 64.

2) Local Structural Feature Interaction Module: Using only global structural features for change detection cannnot identify local change information because some of the small substructures critical to detecting land use changes are difficult to capture in global-level features. Thus, we employed LSIM to extract additional local change information. Specifically, we projected the local node-level structural features S_1 and S_2 of two scene graphs into four vectors q_1, q_2, k_1, and k_2 using independent weighted linear mappings. Subsequently, we calculated two attention matrices $A_1, A_2 \in \mathbb{R}^{H \times H}$ as follows:

$$A_1 = \sigma\left(q_1^{\text{T}} \cdot k_2\right) \tag{10}$$

$$A_2 = \sigma\left(q_2^{\text{T}} \cdot k_1\right). \tag{11}$$

These attention matrices are considered the initial node feature similarity matrices, which are learned during training. By extracting the features of these similarity matrices, we derived the local structural feature interaction results. CNNs are highly effective in extracting high-level features from structured data. However, varying node counts across different scenes is a challenge, which results in the generation of attention matrices of different sizes. Based on [58], we resized the similarity matrices using image resampling techniques. We selected bilinear interpolation for image resampling to ensure that the resulting similarity matrix has a fixed $D \times D$ shape, where D is the hyperparameter controlling information loss

$$R_{A_1} = \text{RES}_D(A_1) \tag{12}$$

$$R_{A_2} = \text{RES}_D(A_2) \tag{13}$$

where R_{A_1} and R_{A_2} are the resized A_1 and A_2, respectively, and RES_D denotes the image resampling operation. The resized matrices are passed through multiple CNNs in parallel to extract the feature representations. The results from each CNN branch are concatenated and fed into multiple fully connected layers, and the final interaction I_{local} for the scene graphs is

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obtained as follows:

$$f(R_{A_1}) = \text{CNNs}(R_{A_1}) \tag{14}$$

$$f(R_{A_1}) = \text{CNNs}(R_{A_1}) \tag{15}$$

$$f_{\text{concat}} = \text{Concat}(f(R_{A_1}), f(R_{A_2})) \tag{16}$$

$$I_{\text{local}} = \text{FC}(f_{\text{concat}}). \tag{17}$$

D. Land Use Change Detection and Loss Function

We concatenated the global structural feature interaction result, I_{global}, with the detailed local structural feature interaction result, I_{local}, to derive an enhanced representation for the scene image pairs, leading to a combined vector, I. This vector is subsequently passed through a fully connected layer to classify the scene images into changed and unchanged categories, yielding the final change detection results

$$I = \text{Concat}(I_{\text{global}}, I_{\text{local}}). \tag{18}$$

1) Loss Function: We used a weighted cross-entropy loss function [59] for land use change detection. This function addresses the common issue of significant imbalance between the numbers of changed and unchanged samples in realistic settings. By assigning a higher weight to the changed class, we enhance the model's sensitivity on these samples, thereby minimizing the negative impacts of sample imbalance. The change detection loss, L, is defined as follows:

$$L = -w_0 y \log \hat{y} - w_1(1-y)\log(1-\hat{y}). \tag{19}$$

On the MtS-WH dataset, where the number of changed and unchanged samples is relatively balanced, we set the loss weights as $w_0 = 1$ and $w_1 = 1.2$. In contrast, the LUCD-FZ and LUCD-HF datasets exhibit a stronger imbalance between changed and unchanged samples, and we raised the value of w_1 to 5.

2) Accuracy Assessment: We evaluated the accuracy of the detected scene-based land use changes using the following metrics: overall accuracy (OA), precision (P), recall (R), and $F1$ score ($F1$). These four metrics are defined as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{20}$$

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{21}$$

$$R = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{22}$$

$$F1 = 2 \times \frac{P \times R}{P+R} \tag{23}$$

where TP, FP, and FN indicate the true positive, false positive, and false negative, respectively. In this context, positive refers to the samples where changes have occurred.

E. Performance Evaluation

1) Comparison of Different Semantic Features: To validate our selection of the high-level semantic features derived from CLCFormer, we conducted comparative experiments on land use change detection, including low- and mid-level semantic features. More specifically, the low-level semantic features encompass spectral features (i.e., spectral mean and standard deviation, brightness, and maximum difference) and texture features (i.e., gray-level co-occurrence matrix mean and standard deviation, contrast, correlation, homogeneity, energy, and entropy). Mid-level semantic features are derived from histogram of oriented gradient (HOG) features.

2) Comparison of Different Structure Feature Extraction Backbones: To evaluate the effectiveness of the adopted GIN as the backbone of the structural feature extractor, we conducted a comparative analysis with several mainstream GNNs, including the GCN [60], the graph sample and aggregate network (GraphSAGE) [61], the graph attention network (GAT) [62], and the graph transformer network (GTN) [63]. Specifically, GCN aggregates node features and neighborhood information through convolutional operations, while GAT uses a self-attention mechanism to learn adaptive weights and capture relationships between nodes. GraphSAGE, in contrast, learns node representations by sampling and aggregating features from neighboring nodes. GTN, employing a transformer-based architecture, processes graph-structured data and captures complex relationships between nodes. Each of these networks, when contrasted with GIN, offers distinct approaches to structural feature extraction, thereby enriching our evaluation and providing insights into their respective strengths.

3) Comparison With Existing Methods for Urban Land Use Change Detection: To evaluate the effectiveness of the proposed method for urban land use change detection, we compared it with two existing methods, corresponding to a traditional method based on BoVW [26] and a CNN-based method, CorrFusionNet [25]. The BoVW-based method uses a BoVW model to encode the bitemporal image scenes and employs a support vector machine classifier to obtain the scene classification results. Scene change detection results are obtained through postclassification comparison. CorrFusionNet employs a CNN to extract deep latent feature representations of bitemporal input image scenes and uses a CorrFusion module to compute the temporal correlations for detecting scene changes. For this study, DenseNet121 was selected as the deep feature extractor for CorrFusionNet.

F. Implementation Details

We applied our method to three scene-based change detection datasets: MtS-WH, LUCD-FZ, and LUCD-HF. The model was trained for 100 epochs using the stochastic gradient descent (SGD) optimizer with a learning rate of 5e-4 and a batch size of 16. Experiments were performed on an NVIDIA GeForce RTX 3080 GPU, and the algorithms were implemented using MATLAB and Python. We used a stratified random sampling technique to ensure an equitable representation of each class within the divided datasets. For each dataset, 60% of the samples were used for training, 20% for validation, and the remaining 20% for testing.

IV. Results

We conducted ablation studies to assess the effectiveness of the GSEM, GSIM, and LSIM. Initially, we established a baseline using SGNN and incrementally introduced each

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TABLE II
ACCURACY EVALUATION OF ABLATION EXPERIMENTS ON THE MTS-WH AND LUCD-FZ DATASETS

Datasets	Groups	LSIM	GSIM	GSEM	OA	P	R	F1	C-F1 ↓
MtS-WH	a (ours)	✓	✓	✓	**0.9476**	**0.9390**	**0.9277**	**0.9333**	
	b	×	✓	✓	0.9333	0.9259	0.9036	0.9146	10.28%
	c	×	×	✓	0.8429	0.7604	0.8795	0.8156	**54.43%**
	d (baseline)	×	×	×	0.7952	0.7222	0.7831	0.7514	35.29%
LUCD-FZ	a (ours)	✓	✓	✓	**0.9870**	**0.9545**	0.8750	**0.9130**	
	b	×	✓	✓	0.9805	0.8214	**0.9583**	0.8846	16.91%
	c	×	×	✓	0.9675	0.7692	0.8333	0.8000	**50.39%**
	d (baseline)	×	×	×	0.9578	0.7037	0.7917	0.7451	32.70%

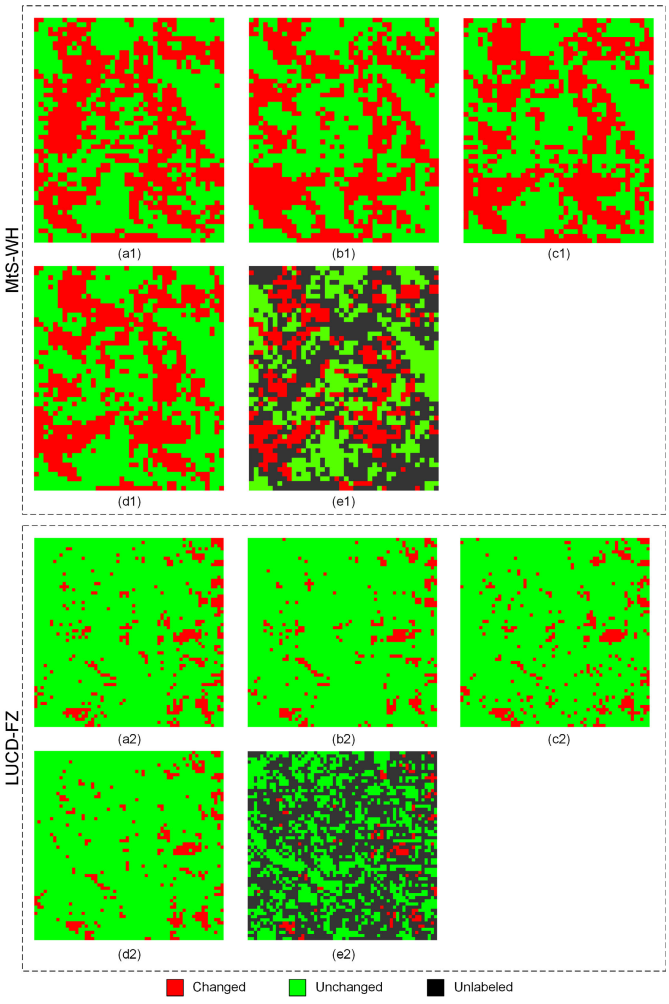


Fig. 6. Land use change detection results using different modules in ablation experiments on the MtS-WH and LUCD-FZ datasets. (a)–(c) Results obtained by the three variants of the method without GSEM, without GSEM and GSIM, and without GSEM, GSIM, and LSIM, respectively. (d) Results obtained by the proposed method. (e) Ground truth.

module, resulting in three model variants. All models use a weighted cross-entropy loss function and were trained using identical hyperparameters and sample conditions to detect changes in land use. Table II lists the accuracy of the ablation experiments on the MtS-WH and LUCD-FZ datasets. Without these modules, a significant reduction in detection accuracy was observed, which indicates that the graph-level interaction module makes the most significant contribution to detection performance. Fig. 6 shows the detection results

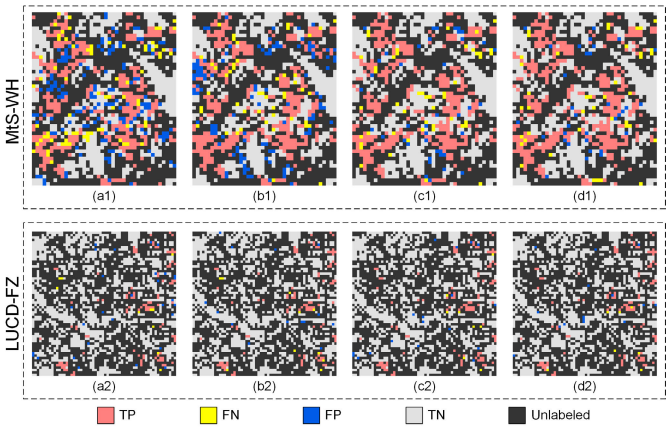


Fig. 7. Distribution of accuracy metrics of the detected results from ablation experiments on the MtS-WH and LUCD-FZ datasets. (a)–(c) Results obtained by the three variants of the method without GSEM, without GSEM and GSIM, and without GSEM, GSIM, and LSIM, respectively. (d) Results obtained by the proposed method.

on both datasets using different modules in the ablation experiments, and Fig. 7 shows the corresponding distribution of the accuracy metrics. Table III lists specific cases where the incorporation of the proposed modules improved detection performance.

A. Comparison of Different Semantic Features

To validate the effectiveness of the high-level semantic features derived from CLCFormer, we compared the detection of changes in land use using low- and mid-level semantic features. Table IV lists the accuracy of these experiments using three levels of semantic features on the MtS-WH and LUCD-FZ datasets. Fig. 8 shows the corresponding detection results for the two datasets, and Fig. 9 shows the distribution of accuracy metrics for the detection results obtained using different semantic features on both datasets. The use of high-level semantic features to detect changes significantly improved the detection results. Specifically, the $F1$ scores increased by 16.41% and 16.32% on the MtS-WH dataset and by 13.04% and 6.85% on the LUCD-FZ dataset.

B. Comparison of Different Structure Feature Extraction Backbones

To evaluate the effectiveness of using GIN as the backbone for structural feature extraction, we conducted comparative experiments with four other mainstream GNNs. The land use

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TABLE III

DEMONSTRATION OF SPECIFIC CASES WHERE INCORPORATING THE PROPOSED MODULES IMPROVED DETECTION PERFORMANCE

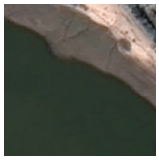
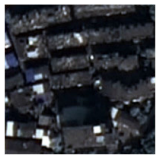
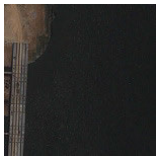
Datasets	**MtS-WH**			**LUCD-FZ**		
Modules	GSEM	GSIM	LSIM	GSEM	GSIM	LSIM
T1 image						
T2 image						
T1 class	Water	Vegetation region	Sparse houses	Water	Green space	High-level residential
T2 class	Water	Vegetation region	Idle region	Water	Green space	High-level residential

TABLE IV

ACCURACY EVALUATION OF COMPARATIVE EXPERIMENTS USING DIFFERENT SEMANTIC FEATURES ON THE MTS-WH AND LUCD-FZ DATASETS

Datasets	**Features**	**OA**	**P**	**R**	**F1**
MtS-WH	Low-level	0.8143	0.7558	0.7831	0.7692
	Mid-level	0.7952	0.6923	0.8675	0.7701
	High-level	**0.9476**	**0.9390**	**0.9277**	**0.9333**
LUCD-FZ	Low-level	0.9675	0.8182	0.7500	0.7826
	Mid-level	0.9773	0.9048	0.7917	0.8445
	High-level	**0.9870**	**0.9545**	**0.8750**	**0.9130**

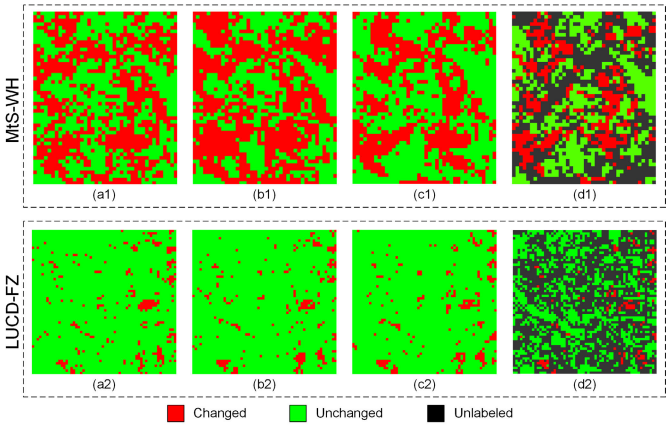


Fig. 8. Land use change detection results using different semantic features on the MtS-WH and LUCD-FZ datasets. (a) Low-level features. (b) Mid-level features. (c) High-level features. (d) Ground truth.

change detection results are presented in Fig. 10. Table V provides the accuracy evaluation of the detection results using five different GNNs on the MtS-WH and LUCD-FZ datasets. Our results show that employing GIN as the backbone for structural feature extraction achieved the best detection performance. Specifically, on the MtS-WH dataset, the $F1$ scores were higher by 2.21%, 1.31%, 7.80%, and 8.27% compared to GCN, GraphSAGE, GAT, and GTN, respectively. On the LUCD-FZ dataset, the $F1$ scores surpassed those of GCN, GraphSAGE, GAT, and GTN by 1.10%, 3.80%, 6.19%, and 5.02%, respectively.

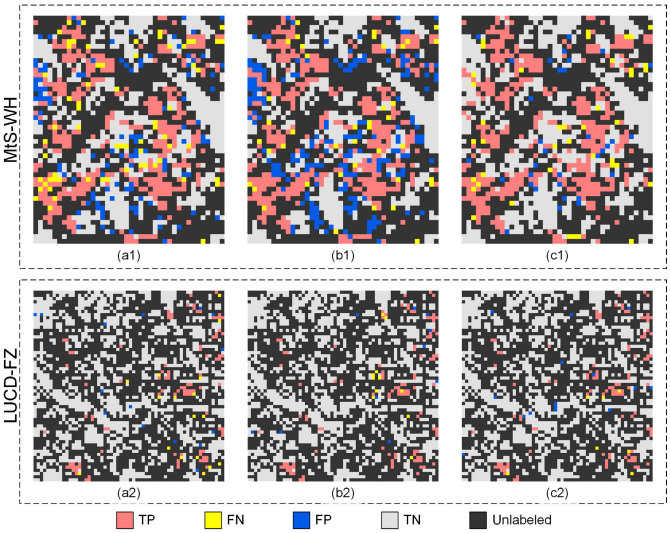


Fig. 9. Distribution of accuracy metrics of the detected results using different semantic features on the MtS-WH and LUCD-FZ datasets. (a) Low-level features. (b) Mid-level features. (c) High-level features.

C. Comparison With Existing Methods for Urban Land Use Change Detection

We compared our method with CorrFusionNet and the BoVW-based method on the MtS-WH and LUCD-FZ datasets. Table VI shows that our method achieved the highest $F1$ scores of 0.9333 on the MtS-WH dataset and 0.9130 on the LUCD-FZ dataset. It improved $F1$ scores by 4.11% and 9.33% over CorrFusionNet and the BoVW-based method on the MtS-WH dataset, and by 27.47% and 27.53% on the LUCD-FZ dataset. These results underscore the effectiveness of our approach in detecting urban land use changes, particularly in imbalanced sample scenarios. Fig. 11 shows the land use change detection results of the three methods on both datasets. The detection results of our method have the highest

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TABLE V

ACCURACY EVALUATION OF COMPARATIVE EXPERIMENTS USING DIFFERENT GNNS ON THE MTS-WH AND LUCD-FZ DATASETS

Datasets	Methods	OA	P	R	F1
MtS-WH	GCN	0.9286	0.8953	**0.9277**	0.9112
	GraphSAGE	0.9381	0.9375	0.9036	0.9202
	GAT	0.8905	0.8947	0.8193	0.8553
	GTN	0.8762	0.8132	0.8916	0.8506
	GIN	**0.9476**	**0.9390**	**0.9277**	**0.9333**
LUCD-FZ	GCN	0.9838	0.8519	**0.9583**	0.9020
	GraphSAGE	0.9805	0.8750	0.8750	0.8750
	GAT	0.9773	0.8696	0.8333	0.8511
	GTN	0.9773	0.8148	0.9167	0.8628
	GIN	**0.9870**	**0.9545**	0.8750	**0.9130**

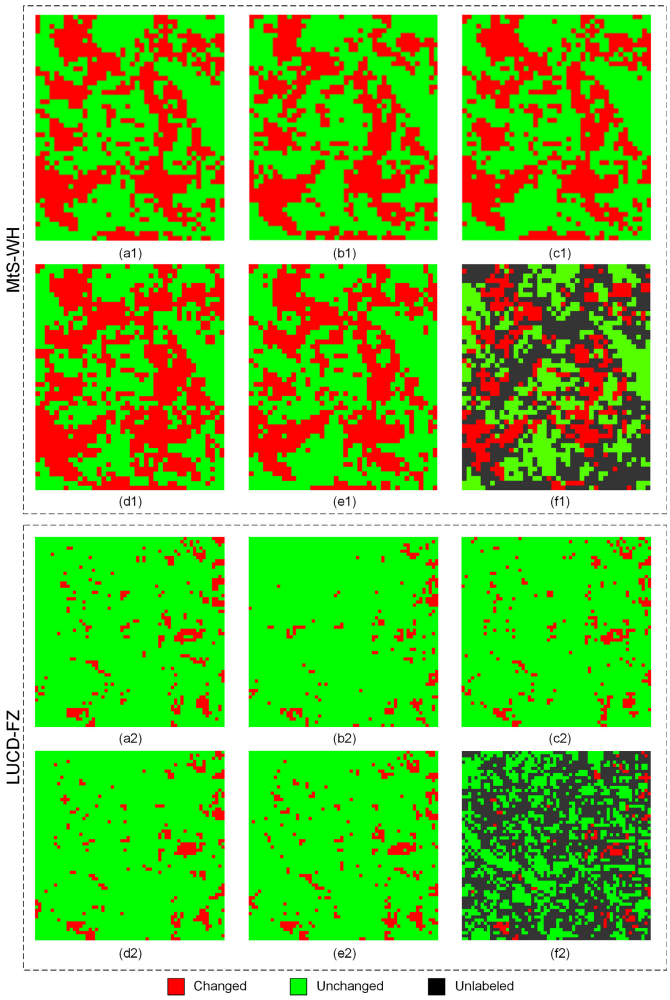


Fig. 10. Land use change detection results using different structural feature extraction networks on the MtS-WH and LUCD-FZ datasets. (a) GCN. (b) GAT. (c) GraphSAGE. (d) GTN. (e) GIN. (f) Ground truth.

agreement with the ground truth, resulting in the fewest false positives and missed detections.

D. Effectiveness of Local–Global Feature Interaction Modules

1) Effectiveness of GSIM: To explore the impact of GSIM on the detection of land use changes, we compared the proposed method with the conventional approach of subtracting

TABLE VI

ACCURACY EVALUATION OF COMPARATIVE EXPERIMENTS BETWEEN DIFFERENT METHODS ON THE MTS-WH AND LUCD-FZ DATASETS

Datasets	Methods	OA	P	R	F1
MtS-WH	BoVW	0.8476	0.7706	0.9231	0.8400
	CorrFusionNet	0.8952	0.9100	0.8750	0.8922
	Ours	**0.9476**	**0.9390**	**0.9277**	**0.9333**
LUCD-FZ	BoVW	0.9191	0.5238	0.8148	0.6377
	CorrFusionNet	0.9094	0.5769	0.7143	0.6383
	Ours	**0.9870**	**0.9545**	**0.8750**	**0.9130**

TABLE VII

ACCURACY EVALUATION OF CHANGE DETECTION RESULTS USING DIFFERENT FEATURE RELATIONSHIP MODELING METHODS

Datasets	Methods	OA	P	R	F1
MtS-WH	Subtraction	0.8810	0.8021	**0.9277**	0.8603
	GSIM	**0.9476**	**0.9390**	**0.9277**	**0.9333**
LUCD-FZ	Subtraction	0.9773	0.8696	0.8333	0.8511
	GSIM	**0.9870**	**0.9545**	**0.8750**	**0.9130**

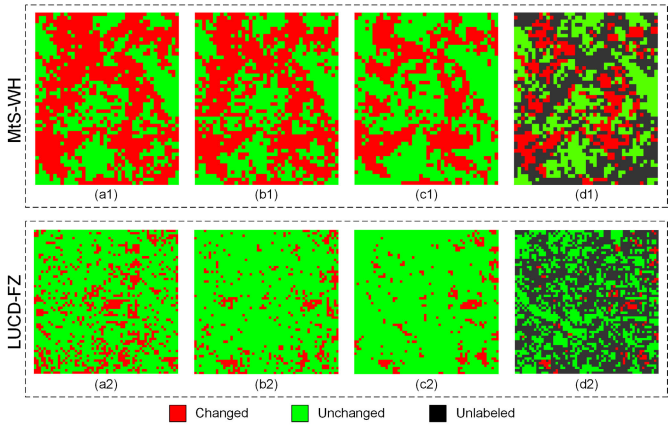


Fig. 11. Land use change detection results by different methods on the MtS-WH and LUCD-FZ datasets. (a) BoVW. (b) CorrFusionNet. (c) Ours. (d) Ground truth.

bitemporal features. Qualitative and quantitative assessments were performed to ensure a comprehensive evaluation of the module. Fig. 12 shows the global structural features derived from the proposed GSIM and the direct bitemporal feature subtraction method. The horizontal axis represents the features of each dimension, and the vertical axis represents the corresponding values. Compared with the direct bitemporal feature subtraction method, the values obtained using GSIM decreased in some channels and increased in others, resulting in larger peak values and a sharper overall distribution. This indicates that the features obtained by GSIM exhibit better discriminative ability. Table VII lists the accuracy of the change detection results obtained by the two methods for the MtS-WH and LUCD-FZ datasets. For both datasets, GSIM significantly improved the detection of land use changes, as indicated by the 7.30% and 6.19% increases in the $F1$ scores on the MtS-WH and LUCD-FZ datasets, respectively.

2) Effectiveness of LSIM: To explore the impact of LSIM on the detection of land use changes, we compared the

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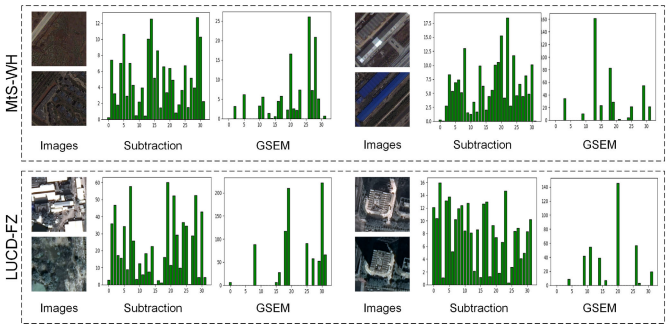


Fig. 12. Comparison of the histograms of global structural features derived from the proposed GSIM module and the direct bitemporal feature subtraction method.

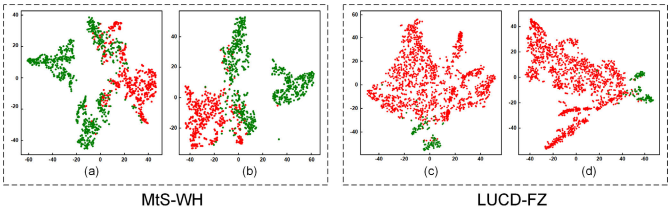


Fig. 13. Comparison of the separability of global structural features enhanced by the LSIM versus those without LSIM on the MtS-WH and LUCD-FZ datasets. (a) and (c) With LSIM. (b) and (d) without LISM.

separability of features derived with and without the LSIM. We employed t-distributed stochastic neighbor embedding to project these high-dimensional features into a 2-D embedding space for separability analysis [64]. Fig. 13 shows the separability of global structural features enhanced by the LSIM versus those without LSIM on the MtS-WH and LUCD-FZ datasets, where green and red points, respectively, represent unchanged and changed scenes. Fig. 13 illustrates enhanced separability between green and red points with the inclusion of the LSIM. Table II lists that integrating the LSIM results in an $F1$ score improvement of 1.87% and 2.84% on the MtS-WH and LUCD-FZ datasets, respectively.

E. Effectiveness of GSEM

To validate the effectiveness of GSEM in generating more representative graph-level features based on node-level features, we compared the detection results with those of traditional mean and maximum aggregation methods. We visualized the attention scores learned for each land cover object in image scenes. Fig. 14(a)–(c) shows the attention scores for different land cover objects from the MtS-WH dataset, and Fig. 14(d)–(f) shows those from the LUCD-FZ dataset. The red and blue regions indicate the nodes that received high and low attention, respectively. These figures demonstrate that the model assigns higher attention to major land cover categories in image scenes and allocates lower attention to minor categories; thus, it focuses on the primary land cover classes and generates more representative global features. Table VIII lists the accuracy of results obtained using different global structural feature generation methods on the MtS-WH and LUCD-FZ datasets. Notably, among all approaches, the proposed GSEM exhibited the highest $F1$ score.

TABLE VIII
ACCURACY EVALUATION OF COMPARATIVE EXPERIMENTS BASED ON DIFFERENT METHODS ON THE MTS-WH AND LUCD-FZ DATASETS

Datasets	Methods	OA	P	R	F1
MtS-WH	Mean	0.8905	0.8409	0.8916	0.8655
	Max	0.8857	0.8642	0.8434	0.8537
	GSEM	**0.9476**	**0.9390**	**0.9277**	**0.9333**
LUCD-FZ	Mean	0.9740	0.8333	0.8333	0.8333
	Max	0.9773	0.8148	**0.9167**	0.8628
	GSEM	**0.9870**	**0.9545**	0.8750	**0.9130**

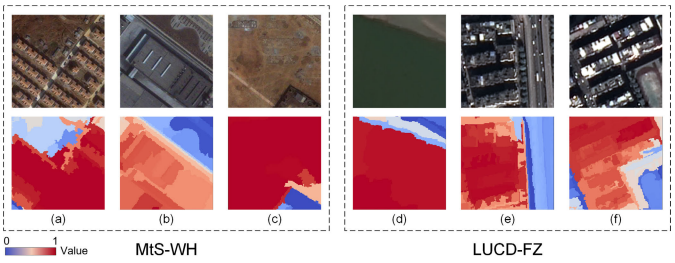


Fig. 14. Examples of attention score values learned for different land cover objects within image scenes. (a)–(c) Examples from MtS-WH. (d)–(f) Examples from LUCD-FZ.

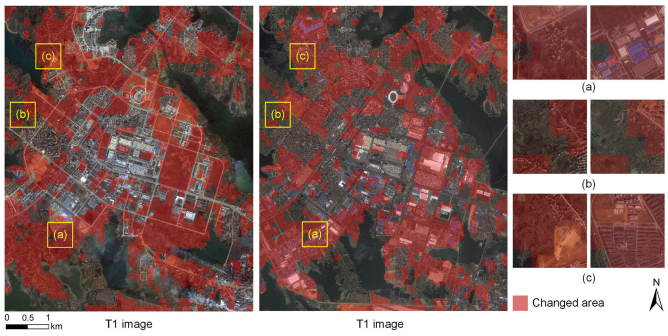


Fig. 15. Land use change detection results by our method on the MtS-WH dataset. (a)–(c) Detection results in specific areas.

F. Land Use Change Detection on Different Datasets

1) Land Use Change Detection on the MtS-WH Dataset: To validate the effectiveness of the proposed land use change detection method, we conducted experiments on different datasets. Fig. 15 shows the land use change detection results on the MtS-WH dataset. The detection results obtained using the proposed method exhibit a high agreement with the actual ground changes. Table IX lists the demonstration of the detection results on the MtS-WH dataset. Significant changes in land use, such as the transformation of dense or sparse houses into residential or industrial regions due to building reconstruction, were accurately detected. In addition, the proposed method was robust against external factors, including changes in the illumination conditions and seasonal variations. For example, in the last three pairs of unchanged land use samples, despite spectral and textural changes due to seasonal or illumination differences, the final detection result remained accurate.

2) Land Use Change Detection on the LUCD-FZ Dataset: Fig. 16 shows the land use change detection results on the LUCD-FZ dataset. The detection results obtained using the

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TABLE IX
DEMONSTRATION OF THE DETECTED RESULTS BY OUR METHOD ON THE MTS-WH DATASET

Dataset	**MtS-WH**					
Type	**Changed**				**Unchanged**	
T1 image						
T2 image						
T1 class	Dense houses	Sparse houses	Sparse houses	Industrial region	Vegetation region	Dense houses
T2 class	Residential region	Industrial region	Residential region	Industrial region	Vegetation region	Dense houses

TABLE X
DEMONSTRATION OF THE DETECTED RESULTS BY OUR METHOD ON THE LUCD-FZ DATASET

Dataset	**LUCD-FZ**					
Type	**Changed**				**Unchanged**	
T1 image						
T2 image						
T1 class	High-density residential	Undeveloped land	Industrial and warehouses	Water	Commercial	Industrial and warehouses
T2 class	Low-density residential	Low-density residential	Undeveloped land	Water	Commercial	Industrial and warehouses

proposed method have a high consistency with the actual ground changes. Table X lists the demonstration of the detection results on the LUCD-FZ dataset. Similar to the findings on the MtS-WH dataset, significant changes in land use were accurately detected. In addition, pseudo-changes caused by differences in illumination were also correctly identified, demonstrating the robustness of the proposed method.

3) Land Use Change Detection on the LUCD-HF Dataset Based on Transfer Learning: To test the transferability of our method to VHR images across different areas, we applied the model trained on the LUCD-FZ dataset to detect land use changes on the LUCD-HF dataset, which also exhibits a significant imbalance between the number of changed and unchanged land use samples. Fig. 17 shows the land use change detection results for the LUCD-HF dataset. Our method successfully detected land use changes during the transfer process, even in images exhibiting significant spectral differences and varied land cover distributions. The performance of the model, when transferred from the LUCD-FZ dataset to the LUCD-HF dataset, was comparable to that achieved by training directly on the LUCD-HF dataset, as indicated by the OA and $F1$ score (Table XI).



Fig. 16. Land use change detection results by our method on the LUCD-FZ dataset. (a)–(c) Detection results in specific areas.

V. DISCUSSION

In this study, we investigated the detection of changes in urban land use from bitemporal VHR satellite images at the

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Fig. 17. Land use change detection results by our method on the LUCD-HF dataset based on transfer learning. (a)–(c) Detection results in specific areas.

TABLE XI
EVALUATION OF DETECTION RESULTS BY TRANSFERRING METHOD TRAINED ON THE LUCD-FZ TO THE LUCD-HF DATASET

Dataset	Trasfer	OA	P	R	F1
LUCD-HF	✓	0.9803	0.9091	0.7692	0.8333
	×	**0.9852**	**0.9167**	**0.8462**	**0.8800**

scene level, focusing on the high-level structural information inherent in scene images. The results demonstrate that the proposed method is effective in detecting changes in land use in various urban areas and the high-level semantic information extracted by the pretrained model improves the detection results. The adopted GIN-derived structural features outperformed other GNN methods in detecting changes in urban land use. Furthermore, the integration of global and local structural feature interactions minimized the false detection caused by changes in illuminance conditions and seasonal variations, as well as missed detection resulting from local structural variations in complex scenarios. Most existing scene-based change detection methods focus on extracting high-level semantic features from VHR images, but only a few studies have explored the incorporation and interaction of bitemporal structural information for detecting changes in urban scene-based land use from VHR images. In this study, we developed an LGSI-Net model, which incorporates the learning of high-level structural features and bitemporal feature interactions to address this gap.

A. Contribution of Advanced Semantic Information Extraction to the Detection of Land Use Change

We represented bitemporal VHR scene images as graph-structure data, with land cover objects as nodes and their connections as edges. We designed a set of handcrafted features, including texture and spectral features, to capture the semantic information of land cover objects from bitemporal scene images. This traditional method, which is commonly used in various applications, did not yield satisfactory results. Thus, we explored the extraction of mid-level vision features using HOG at a higher level, but these features also failed to produce satisfactory results. To address this, we adopted a more advanced semantic feature extraction method based on deep learning. Specifically, we used CLCFormer, a robust semantic segmentation model, to extract high-level semantic features for nodes. Table IV lists the change detection accuracy based on different semantic features on the MtS-WH and LUCD-FZ datasets. A significant improvement in accuracy was observed using high-level semantic features compared to low- and mid-level features. Although the deep learning-based feature extraction approach yielded satisfactory results, training a deep learning model for feature extraction requires sufficient samples, which limits its applications. Further studies are required to explore more advanced methods for extracting deep semantic features from images using semisupervised or unsupervised approaches.

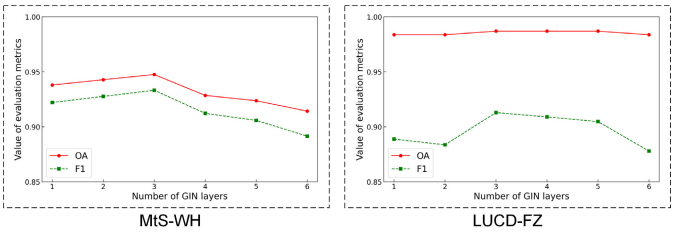


Fig. 18. Effect of the GNN layer depth on land use change detection metrics on the MtS-WH and LUCD-FZ datasets.

B. Contribution of Advanced Structure Feature Extractor to the Detection of Land Use Change

We used three-layer GIN as the backbone for our structural feature extractor to derive high-level structural features from bitemporal scene graphs, facilitating the quantification of the spatial structure information. GIN has been used for various learning tasks on graph-structured data. However, it has not been explored in extracting structural differences in land use mapping or change detection. Previous studies have investigated the effectiveness of GCN for feature learning in land use mapping [1], [40]. To demonstrate the effectiveness of GIN as the backbone of the land use structural feature extractor, we compared it with GCN and three other mainstream GNNs. The results show that GIN exhibited higher accuracy than the other methods in detecting changes in land use. Considering the rapid developments in deep learning methodologies, more advanced structural feature extractors can still be explored to improve the detection performance. To assess the influence of the GNN layer depth on the effectiveness of structural feature extraction for the detection of land use changes, we conducted a sensitivity analysis. The results, shown in Fig. 18, indicate that a three-layer GNN configuration yields the best performance on both the MtS-WH and LUCD-FZ datasets.

C. Effectiveness of Local–Global Structural Feature Interaction Modules

We designed GSIM and LSIM to capture the temporal correlation of bitemporal scene features. Recent studies have attempted to minimize false detection of changes in land use through bitemporal feature interactions [49], [50]. Herein, we leverage the advantages of GSIM in modeling relationships between vectors to minimize such false detection. The results show that GSIM enhances certain informative channels and disregards irrelevant ones, thus facilitating the generation of more discriminative interaction results, effectively minimizing

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false detection due to changes in illuminance conditions and seasonal variations. Recent studies have also explored the use of the local change information [31]. Herein, we designed LSIM to capture detailed change information at the level of land cover objects, complementing fine-grained structural change information. This module helps to minimize the potential of missed detection of land use changes resulting from local structural variations in complex scenarios. Ablation experiments demonstrated that incorporating LSIM improved the accuracy of global scene-based change detection, with the $F1$ score increasing from 0.9146 to 0.9333 and from 0.8846 to 0.9130, respectively (Table II). Moreover, incorporating the LSIM module helped detect land use changes caused by subtle structural differences, such as a scene changing from "Sparse houses" to "Idle region" where only a few small buildings were demolished (Table III).

D. Contributions of High-Level Semantic Features and Structural Feature Interactions to the Detection of Land Use Change

In this study, we found that both the high-level semantic features derived from CLCFormer and the proposed structural feature interaction modules (GSIM and LSIM) improve the detection of land use changes, though the extent of improvement varies between datasets. Ablation and comparison experiments (Tables II and IV) showed that incorporating local and global interaction modules improved the $F1$ scores by 11.77% and 11.30% on the MtS-WH and LUCD-FZ datasets, respectively. Compared to mid-level features, high-level semantic features improved the $F1$ scores by 16.32% on the MtS-WH dataset and by 6.85% on the LUCD-FZ dataset. The LUCD-FZ dataset is more complex, characterized by an imbalanced distribution of changed and unchanged samples, making land use change detection more challenging. In such scenarios, the enhancement of structural differences using the proposed modules leads to a significant improvement in detection accuracy.

E. Effect of Spatial Units on Land Use Change Detection

While using scenes as the basic spatial unit for land use analysis is efficient and convenient, as demonstrated in many land use classifications [52], this approach inevitably introduces geometric uncertainties due to detected boundary inconsistencies with ground-truth data. Future work could explore using street blocks or official land use boundaries for land use change detection. In addition, developing advanced methods to automatically delineate land use units from high-resolution remote sensing images may further improve the geometric accuracy of detected changes and extend the proposed method's applicability. In this study, we performed land use change detection based on fixed-sized scenes of 200×200 pixels, following existing studies [27], [39], [52]. This approach yielded satisfactory detection results. However, since ground features vary in size, using an adaptive determination method or analyzing at multiple scales may further improve detection accuracy [65].

F. Model Applicability

We evaluated the effectiveness of the proposed method in detecting changes in urban land use using the MtS-WH and LUCD-FZ datasets. The LUCD-FZ dataset was created for this study due to the limited availability of public datasets. The MtS-WH dataset exhibited a high sample balance, resulting in excellent land use change detection. In contrast, the LUCD-FZ dataset demonstrated a significant sample imbalance, with fewer changed samples than unchanged ones. The detection of land use changes was also more challenging. The proposed method can be applied to both scenarios. In addition, we performed a transferability test by transferring the proposed method trained on the LUCD-FZ samples to detect land use changes on the LUCD-HF dataset. The proposed framework exhibited high transferability, promoting its wide application potential. Moreover, the two created datasets facilitate the development and evaluation of more advanced methods for land use change detection. This study focuses on man-made land use classes such as residential, commercial, and industrial land in urban areas, which account for most of the urban areas. Therefore, we grouped natural land use classes such as forest, orchard, and grassland into a single category, i.e., green space. Detailed land use classes referring to public services are difficult to recognize accurately using remote sensing images alone and occupy small changed areas, making it challenging to create datasets with sufficient samples. Future efforts can be made to enrich these datasets to include more detailed land use classes.

VI. Conclusion

In this study, we developed a method for detecting urban scene-based land use changes from bitemporal VHR images incorporating structural information. We represented the spatial relationship between land cover objects as graphs and constructed a weight-sharing SGNN equipped with an attention-based GSEM to extract high-level structural features inherent in scene image pairs. To minimize false and missed detections caused by external conditions and minor structural changes, thereby improving the accuracy of detecting land use changes, we modeled structural feature interactions at both global and local levels using GSIM and LSIM. We investigated urban land use changes using three datasets: MtS-WH, LUCD-FZ, and LUCD-HF. Based on the obtained results, we deduce the following conclusions.

1) The proposed method outperforms existing models (e.g., CorrFusionNet and BoVW-based method) in detecting land use changes on both MtS-WH and LUCD-FZ datasets, with an $F1$ score of up to 0.9333 on the MtS-WH dataset.
2) Compared to low- and mid-level semantic features, the use of high-level semantic features improves the change detection accuracy on both the MtS-WH and LUCD-FZ datasets. The GIN-derived structural features outperformed other GNN-derived features.
3) The integration of a GSIM minimizes false detections caused by pseudo-changes, and the LSIM minimizes

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false and missed detections caused by local structural changes.

4) The proposed detection method exhibits promising spatial transferability when applied to different study areas, achieving an $F1$ score of up to 0.8333 on the LUCD-HF dataset.

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